



ELECTRO MAGNETI C FI ELD THEORY(EMFT)

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ELECTRO MAGNETI C FI ELD

THEORY

El ect r o s t a t i c s

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Text Book

1. “Principles of Electromagnetics”,
Matthew N. O. Sadi ku,
4th / 6th Edition, Oxford University
Press, 2007 / 2018.

Reference Books:

2. “Electromagnetic Field Theory
Fundamentals”, Bagh Singh Guru, Huseyin
R. Hiziroglu, Cambridge University
Press, 2nd Edition, 2002.
2. “Engineering Electromagnetics”,
William H. Hayt Jr, J. A. Buck, 7th
Edition, Tata McGraw Hill, 2007.
3. “Electromagnetics - Schaum's
Outlines”, J. A. Edminister, Second
Edition, TMH, 2006.
4. “Engineering Electromagnetics

EMF Theory – Unit – 1 { Text Book Sec. 4.1 - 4.10 & 3.4 – 3.7 }

❖ Coulomb's Law and Field Intensity.

❖ Electric Fields due to Continuous Charge Distributions.

❖ Electric Flux Density.

❖ Gauss's Law & Maxwell's Equations.

❖ Divergence of a Vector & Divergence Theorem

❖ Applications of Gauss's Law

❖ Electric Potential.

❖ Relationship between and – Maxwell's



EMF Theory – Unit – 1 { Text Book Sec. 4.1 - 4.8, 5.8, 5.9, 6.2 & 6.4 }



- ❖ Curl of a Vector & Stokes Theorem
- ❖ An Electric Dipole.
- ❖ Energy Density in Electrostatic Fields.
- ❖ Electrostatic Energy.

ELECTRO MAGNETI C F I ELD

THEORY

Coul omb' s Law

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EMF THEORY – Unit - 1

COULOMB'S LAW & FIELD INTENSITY



- Electrostatics – Concepts that are applicable to static (or **time invariant**) electric fields in free space (or vacuum).
- An electrostatic field is produced by '**static**' charge distribution.
- Governed by two fundamental experimental laws: **Coulomb's Law** and **Gauss's Law**.
- **Charles Augustin de Coulomb (1736 - 1806)**, a French Physicist. Published 7 papers in electricity and magnetism

- The unit of electrostatic charge is **Coulomb (C)**

COULOMB'S LAW



- This experimental law was formulated in 1785, deals with the force a point charge exerts on another point charge.
- The law states that the force (F) between two point charges q_1 and q_2 is;
 - Along the line joining them
 - Directly proportional to the product of the charges.
 - Inversely proportional to the square of the distance (R) between them

▪ Similar to Newton's law of gravitation

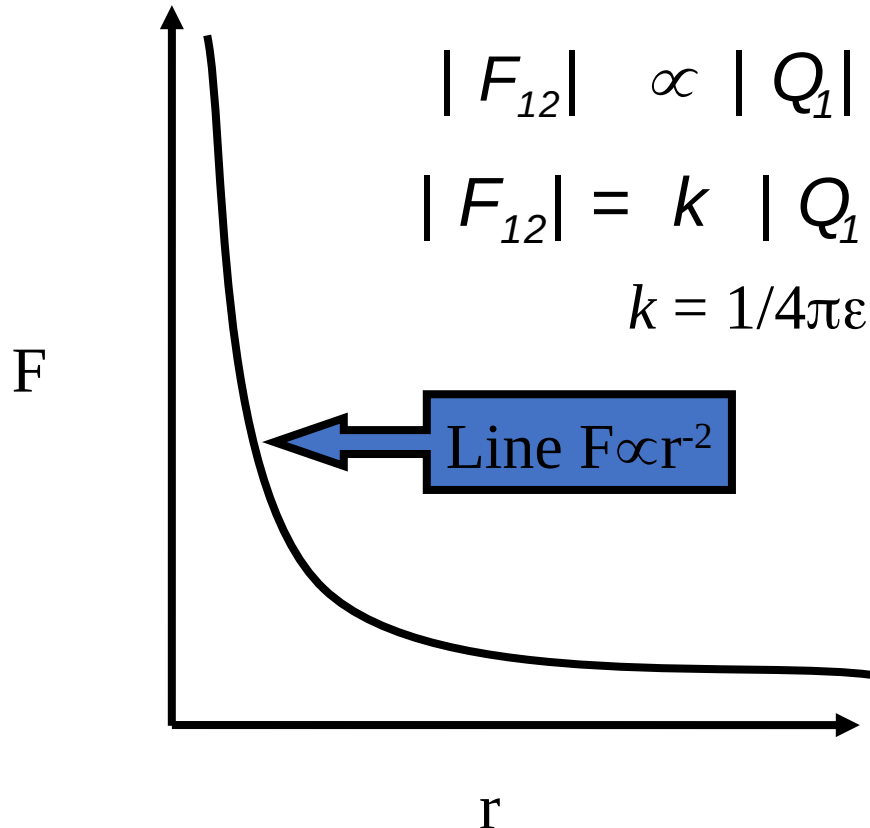
COULOMB'S LAW



- Mathematically, this law can be expressed as;
$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$
- ϵ_0 – constant of proportionality.
- In SI system charges in Coulombs (C), distance (R) in metres (m) and force (F) in Newtons (N).
- $\epsilon_0 = 8.85 \times 10^{-12}$ C²/N m² Farads (m/F)
- ϵ_0 (F/m) – permittivity of free space

ELECTRO MAGNETIC THEORY

Coulomb's Law Experimental Result



$$|F_{12}| \propto |Q_1| |Q_2| / r_{12}^2$$

$$|F_{12}| = k |Q_1| |Q_2| / r_{12}^2$$

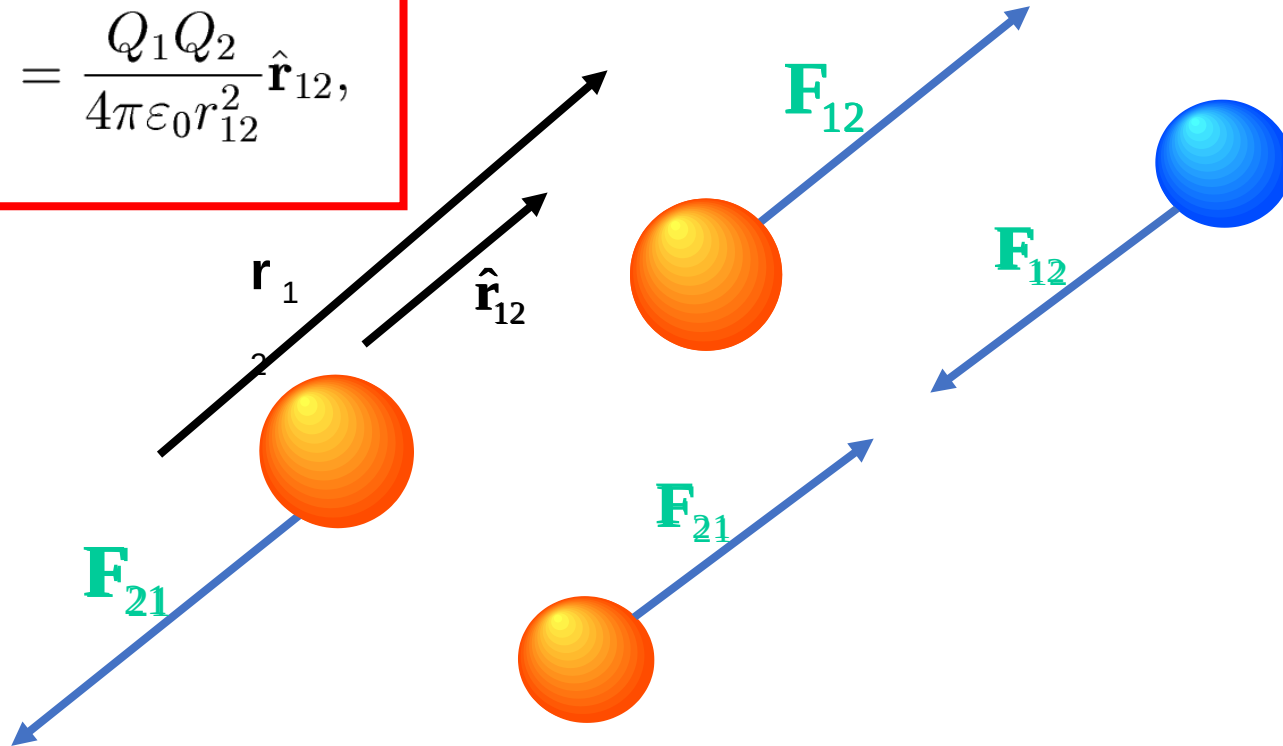
$$k = 1/4\pi\epsilon_0 = 9 * 10^9$$

ELECTRO MAGNETIC THEORY

Coulomb's Law in Vector

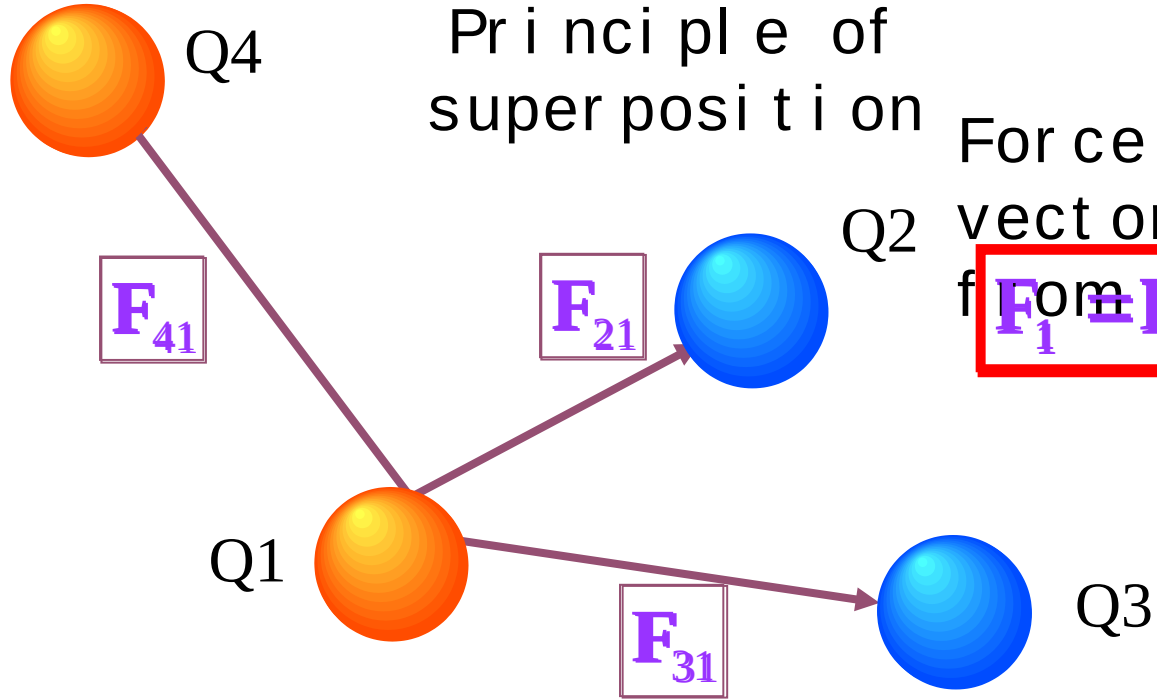
Form

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r_{12}^2} \hat{\mathbf{r}}_{12},$$



ELECTRO MAGNETIC THEORY

Force on many charges



Principle of
superposition

Force on charge is
vector sum of forces
from all charges

$$\mathbf{F}_1 = \mathbf{F}_{21} + \mathbf{F}_{31} + \mathbf{F}_{41}$$

ELECTRO MAGNETIC THEORY

Coulomb's Law V/s Newton's Law of Gravitation

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{|\mathbf{r}_{12}|^2} \hat{\mathbf{r}}_{12}$$

$$\mathbf{F}_{12} = -G \frac{m_1 m_2}{|\mathbf{r}_{12}|^2} \hat{\mathbf{r}}_{12}$$

- Attractive or repulsive
- $1/r^2$
- very strong
- only relevant relatively local scales

Always attractive

$1/r^2$

very weak

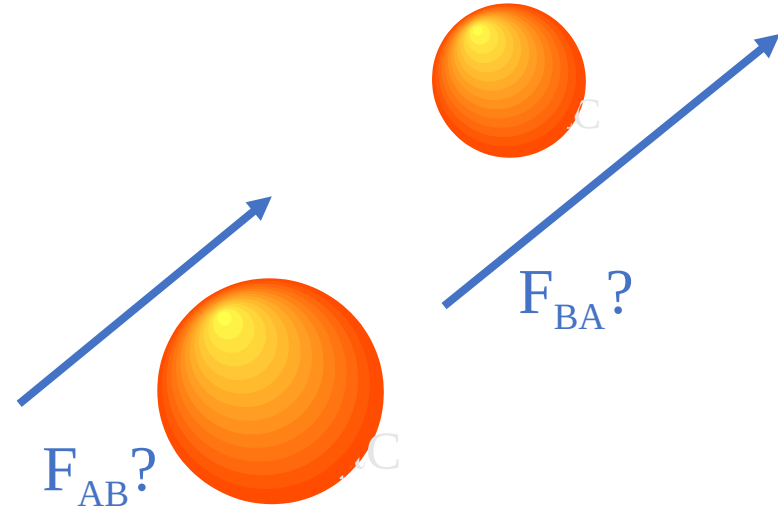
$$\frac{e^2}{4\pi\epsilon_0} \gg -Gm^2$$

important on very large scales,
planets, the Universe

ELECTRO MAGNETI C F i e l d THEORY

QUIZ

Object A has a charge of $+2 \mu\text{C}$ and Object B has a charge of $+6 \mu\text{C}$. Which statement is true?



- **A:** $\mathbf{F_{AB} = -3F_{BA}}$
- **B:** $\mathbf{F_{AB} = -F_{BA}}$
- **C:** $\mathbf{3F_{AB} = -F_{BA}}$
- **D:** $\mathbf{F_{AB} = 12F_{BA}}$

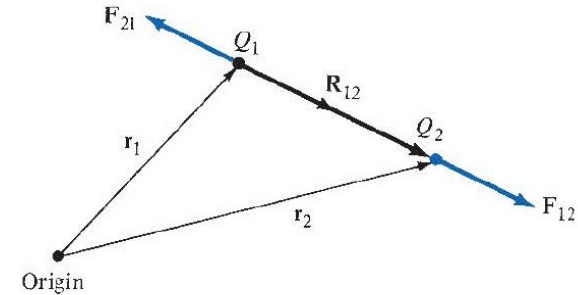
ELECTRO MAGNETI C F I E L D

THEORY Coul omb' s Law and El ect r i c F i e l d I n t e n s i t y E x a m p l e s

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- If point charges are located at points having position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, then the force on Q_1 due to Q_2 is;
- $\hat{\mathbf{r}}_{12}$ - unit vector along $\mathbf{r}_{12}$.
- where $k = \frac{1}{4\pi\epsilon_0}$ and ϵ_0 is the permittivity of free space.
- Substituting for $\hat{\mathbf{r}}_{12}$ in (1), we have;

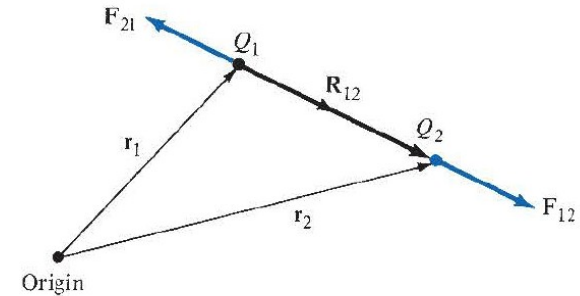


EMF THEORY – Unit - 1

COULOMB'S LAW



- The force on due to is;
- .
- Like charges (charges of same sign) repel each other, while unlike charges attract.
- The distance R between the charged bodies and must be large compared to the linear dimensions of the bodies - and must be point charges !
- and must be static (at rest).
- The signs of and must be taken into



- Principle of Superposition: The principle states that if there are charges located, respectively, at points with position vectors, the resultant force on a charge located at a point with its position vector, is the **vector sum** of the forces exerted on by each of the charges.
- Hence,
-

- The electric field intensity or electric field strength (E) is the force that a unit positive charge experiences when placed in an electric field.
- Hence, $E = \frac{F}{q}$; Newt ons/ Coul omb or vol t s/ met r e.
- The electric field intensity at a point due to a point charge located at r can be expressed as;

- For charges located, respectively, at points with position vectors, the electric field intensity at a point with its position vector, is obtained from **Superposition Principle** as;
-
- Some Numerical.

ELECTRO MAGNETIC FIELD THEORY

Assignment



1. A charge of -10 nC is placed at origin and a charge $Q \text{ C}$ is placed at $(2,0,0)$. If X component of Electric field E is Zero at $(3,1,1)$, Then find Q .
2. Point charges 5 nC and -2 nC are located at $(2,0, 4)$ and $(-3,0, 5)$, respectively.
 - (a) Determine the force on a 1 nC point charge located at $(1, -3, 7)$.
 - (b) Find the electric field E at $(1, -3, 7)$

ELECTRO MAGNETIC FIELD THEORY

Assignment



3. Three equal point charges of $2\mu\text{C}$ are placed at $(0,0,0)$, $(2,0,0)$ and $(0,2,0)$. Find the electrostatic Force on $5\mu\text{C}$ placed at $(2,2,0)$.

ELECTRO MAGNETI C F i e l d THEORY

Assi gnment



Point charges $Q_1 = 5 \mu\text{C}$ and $Q_2 = -4 \mu\text{C}$ are placed at $(3, 2, 1)$ and $(-4, 0, 6)$, respectively. Determine the force on Q_1 .

Five identical $15\text{-}\mu\text{C}$ point charges are located at the center and corners of a square defined by $-1 < x, y < 1, z = 0$.

- (a) Find the force on the $10\text{-}\mu\text{C}$ point charge at $(0, 0, 2)$.
- (b) Calculate the electric field intensity at $(0, 0, 2)$.

Point charges Q_1 and Q_2 are, respectively, located at $(4, 0, -3)$ and $(2, 0, 1)$. If $Q_2 = 4 \text{ nC}$, find Q_1 such that

- (a) The $\mathbf{E}$ at $(5, 0, 6)$ has no z -component
- (b) The force on a test charge at $(5, 0, 6)$ has no x -component.

Charges $+Q$ and $+3Q$ are separated by a distance 2 m . A third charge is located such that the electrostatic system is in equilibrium. Find the location and the value of the third charge in terms of Q .

ELECTRO MAGNETI C

THEORY

St oke's Theor em, Di ver gence

Theor em

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ELECTRO MAGNETI C THEORY

St oke' s Theor em



St oke' s theor em states that Line
i ntegr at i on of any Vect or A i s
equal t o sur face i ntegr at i on of
curl of the same vect or .

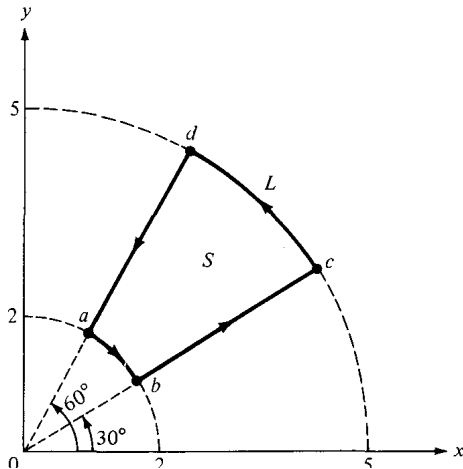
$$= \oint_C \mathbf{A} \cdot d\mathbf{s}$$

St oke' s theor em gi ves the
r el at i onshi p bet ween
Li ne i ntegr at i on t o sur face
i ntegr at i on

ELECTRO MAGNETI C THEORY

St oke' s Theor em

If $\mathbf{A} = \rho \cos \phi \mathbf{a}_\rho + \sin \phi \mathbf{a}_\phi$, evaluate $\oint \mathbf{A} \cdot d\mathbf{l}$ around the path shown in Figure
Confirm this using Stokes's theorem.



Figure

ELECTRO MAGNETI C THEORY

St oke' s Theor em



Solution:

Let

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \left[\int_a^b + \int_b^c + \int_c^d + \int_d^a \right] \mathbf{A} \cdot d\mathbf{l}$$

where path L has been divided into segments ab , bc , cd , and da as in Figure 3.22.

Along ab , $\rho = 2$ and $d\mathbf{l} = \rho d\phi \mathbf{a}_\phi$. Hence,

$$\int_a^b \mathbf{A} \cdot d\mathbf{l} = \int_{\phi=60^\circ}^{30^\circ} \rho \sin \phi d\phi = 2(-\cos \phi) \Big|_{60^\circ}^{30^\circ} = -(\sqrt{3} - 1)$$

ELECTRO MAGNETIC THEORY

Stoke's Theorem



Along bc , $\phi = 30^\circ$ and $d\mathbf{l} = d\rho \mathbf{a}_\rho$. Hence,

$$\int_b^c \mathbf{A} \cdot d\mathbf{l} = \int_{\rho=2}^5 \rho \cos \phi d\rho = \cos 30^\circ \frac{\rho^2}{2} \Big|_2^5 = \frac{21\sqrt{3}}{4}$$

Along cd , $\rho = 5$ and $d\mathbf{l} = \rho d\phi \mathbf{a}_\phi$. Hence,

$$\int_c^d \mathbf{A} \cdot d\mathbf{l} = \int_{\phi=30^\circ}^{60^\circ} \rho \sin \phi d\phi = 5(-\cos \phi) \Big|_{30^\circ}^{60^\circ} = \frac{5}{2}(\sqrt{3} - 1)$$

ELECTRO MAGNETIC THEORY

Stoke's Theorem



Along da , $\phi = 60^\circ$ and $d\mathbf{l} = d\rho \mathbf{a}_\rho$. Hence,

$$\int_d^a \mathbf{A} \cdot d\mathbf{l} = \int_{\rho=5}^2 \rho \cos \phi d\rho = \cos 60^\circ \frac{\rho^2}{2} \Big|_5^2 = -\frac{21}{4}$$

Putting all these together results in

$$\begin{aligned} \oint_L \mathbf{A} \cdot d\mathbf{l} &= -\sqrt{3} + 1 + \frac{21\sqrt{3}}{4} + \frac{5\sqrt{3}}{2} - \frac{5}{2} - \frac{21}{4} \\ &= \frac{27}{4}(\sqrt{3} - 1) = 4.941 \end{aligned}$$

ELECTRO MAGNETIC THEORY

Stoke's Theorem



Using Stokes's theorem (because L is a closed path)

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

But $d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$ and

$$\begin{aligned} \nabla \times \mathbf{A} &= \mathbf{a}_\rho \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] + \mathbf{a}_\phi \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] + \mathbf{a}_z \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right] \\ &= (0 - 0)\mathbf{a}_\rho + (0 - 0)\mathbf{a}_\phi + \frac{1}{\rho} (1 + \rho) \sin \phi \mathbf{a}_z \end{aligned}$$

ELECTRO MAGNETIC THEORY

Stoke's Theorem



Hence:

$$\begin{aligned}\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} &= \int_{\phi=30^\circ}^{60^\circ} \int_{\rho=2}^5 \frac{1}{\rho} (1 + \rho) \sin \phi \rho d\rho d\phi \\ &= \int_{30^\circ}^{60^\circ} \sin \phi d\phi \int_2^5 (1 + \rho) d\rho \\ &= -\cos \phi \bigg|_{30^\circ}^{60^\circ} \left(\rho + \frac{\rho^2}{2} \right) \bigg|_2^5 \\ &= \frac{27}{4} (\sqrt{3} - 1) = 4.941\end{aligned}$$

ELECTRO MAGNETIC THEORY

Divergence Theorem



Divergence Theorem states that surface integration of any vector is same as divergence of the same vector in volume integration.

=

=

Divergence Theorem gives the relationship between Surface integration to volume integration

ELECTRO MAGNETIC THEORY

Divergence Theorem

(Assignment)

Verify the divergence theorem



$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dv$$

for each of the following cases:

- (a) $\mathbf{A} = xy^2\mathbf{a}_x + y^3\mathbf{a}_y + y^2z\mathbf{a}_z$ and S is the surface of the cuboid defined by $0 < x < 1$, $0 < y < 1$, $0 < z < 1$
- (b) $\mathbf{A} = 2\rho z\mathbf{a}_\rho + 3z \sin \phi \mathbf{a}_\phi - 4\rho \cos \phi \mathbf{a}_z$ and S is the surface of the wedge $0 < \rho < 2$, $0 < \phi < 45^\circ$, $0 < z < 5$
- (c) $\mathbf{A} = r^2\mathbf{a}_r + r \sin \theta \cos \phi \mathbf{a}_\theta$ and S is the surface of a quarter of a sphere defined by $0 < r < 3$, $0 < \phi < \pi/2$, $0 < \theta < \pi/2$.

ELECTRO MAGNETI C

THEORY

Gauss Law and i t s Appl i cat i ons

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ELECTRO MAGNETIC THEORY

Electric Field due to continuous charge distributions

Q
+ ●

Point
charge

ρ_L
+ + + +

Line
charge

+ + +
 ρ_S
+ + +

Surface
charge

+ + +
+ + +
 ρ_V
+ + +

Volume
charge

ELECTRO MAGNETIC THEORY



Electric Field due to continuous charge

distributions Contd.,

$$dQ = \rho_L dl \rightarrow Q = \int_L \rho_L dl \quad (\text{line charge})$$

$$dQ = \rho_S dS \rightarrow Q = \int_S \rho_S dS \quad (\text{surface charge})$$

$$dQ = \rho_v dv \rightarrow Q = \int_v \rho_v dv \quad (\text{volume charge})$$

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

ELECTRO MAGNETI C THEORY

El ect r i c Fi el d due t o cont i nuous
char ge di st r i but i ons Cont d. ,



$E =$ Li ne Char ge

a_R Sur f ace Char ge

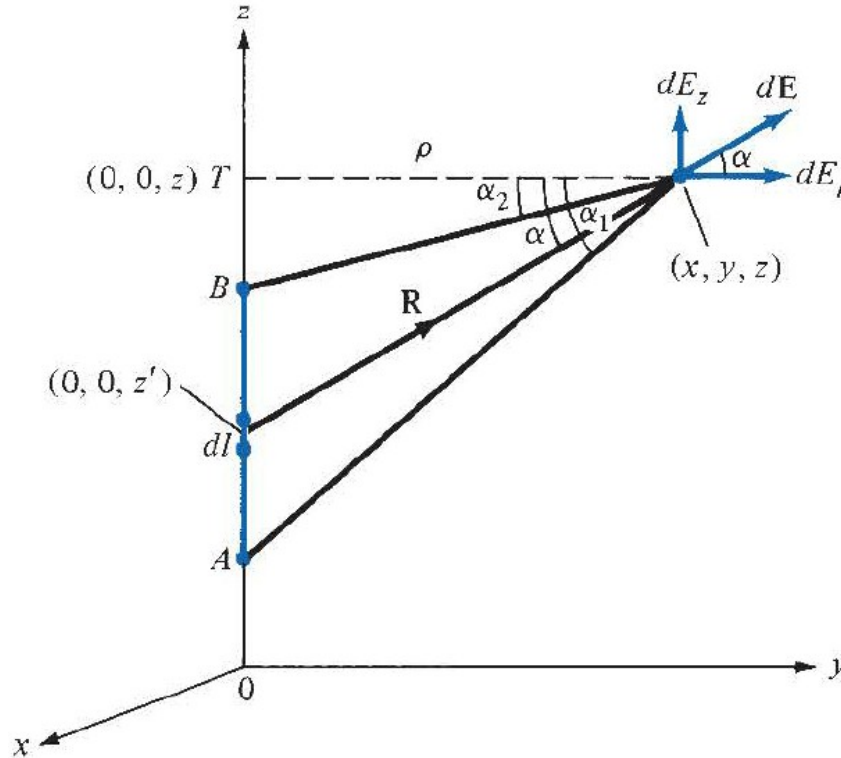
a_R Vol ume Char ge

EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A LINE CHARGE

- Consider a line charge with uniform charge density extending from A to B along the z -axis as shown.

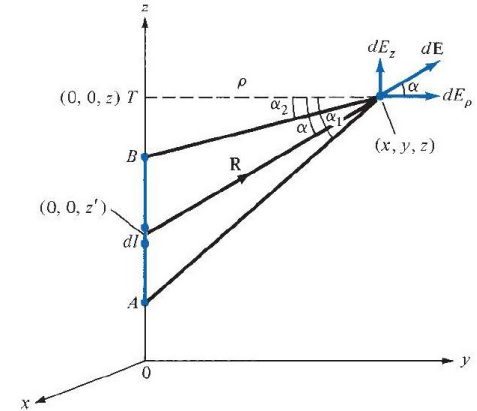
- The charge element associated with the element of the line is,



EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A LINE CHARGE

- The electric field intensity at any arbitrary point can be found using the equation;
- The point at which the field needs to be evaluated is denoted as **field point** as and the **source point** as .
- Hence and
- In cylindrical form - Why ??

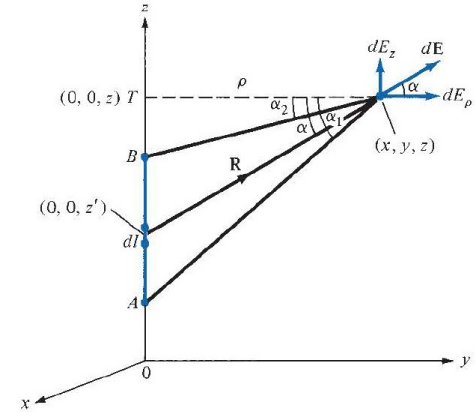


EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A LINE CHARGE



- Further
- Substituting the above in the expression for , we get,
- To evaluate the above integral , we use
and
- , and



EMF THEORY – Unit - 1

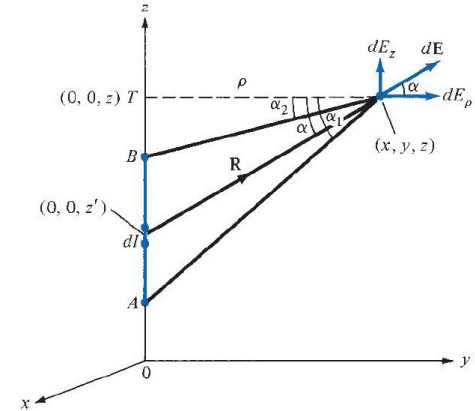
ELECTRIC FIELD DUE TO A LINE CHARGE

- Then the expression for becomes;

How ???

- For Finite Line Charges (with and are known !)

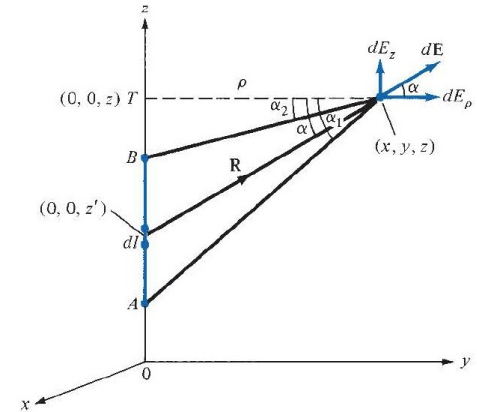
- For an Infinite Line Charge, and , and , the z – component vanishes as



EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A LINE CHARGE

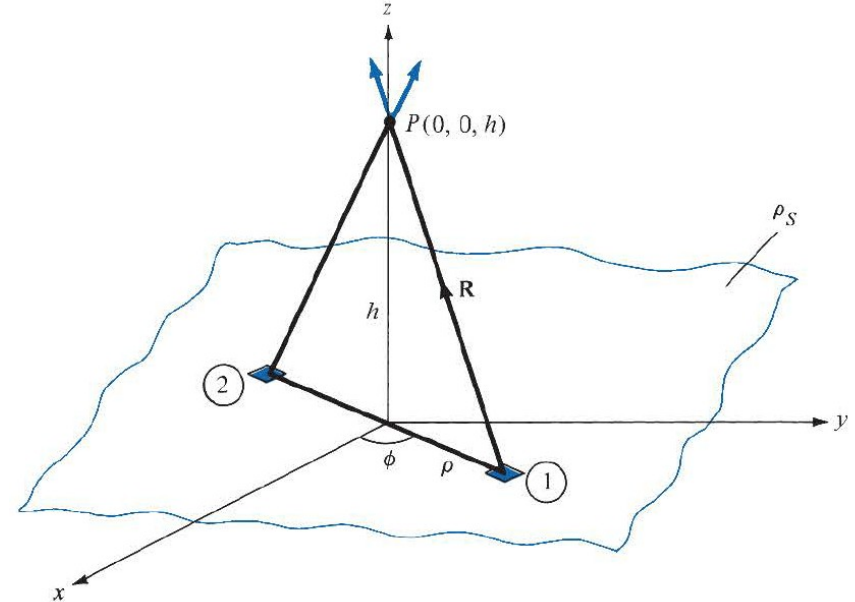
- Then for an infinite line charge;
- In this case, the infinite line charge is along the z – axis so that and have their usual meaning.
- If the line is not along the z – axis,
is the perpendicular distance from this line to the point of interest.
- is a unit vector along that distance directed from the line charge to the ‘field point’.



EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A SURFACE CHARGE

- Consider an infinite sheet of charge in the xy -plane with uniform charge density ρ_s .
- The charge element associated with the elemental area is,
- The electric field intensity due to a surface charge is,



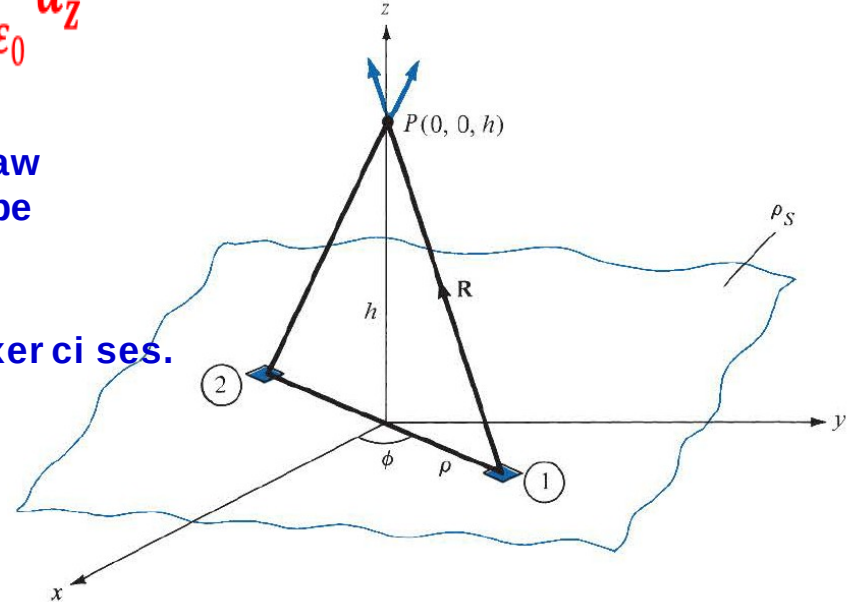
EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A SURFACE CHARGE

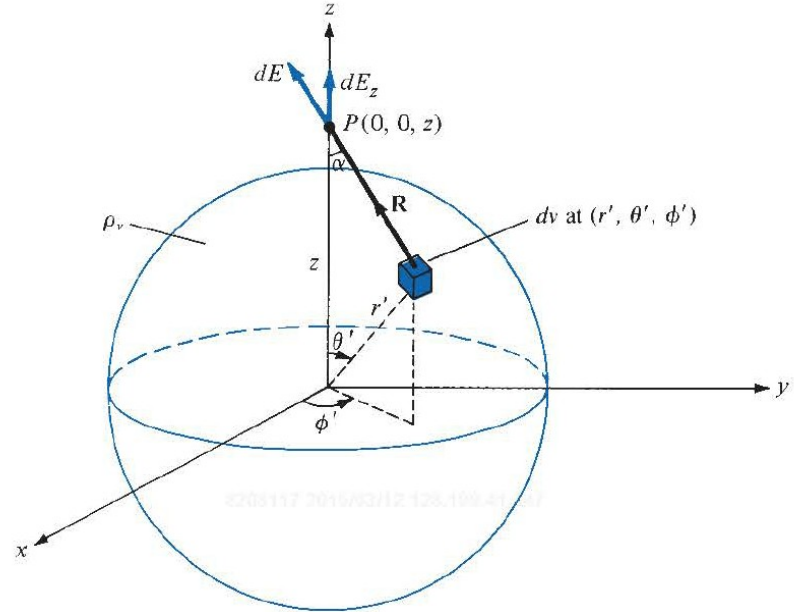
$$E = \int_S dE_z = \frac{\rho_S}{4\pi\epsilon_0} \int_{\varphi=0}^{2\pi} \int_{\rho=0}^{\infty} \frac{h\rho d\varphi d\rho}{[\rho^2 + h^2]^{\frac{3}{2}}} a_z = \frac{\rho_S}{2\epsilon_0} a_z$$

- The proof can be done in Gauss law applications. Here Formula to be known no derivation.

re Numerical. Text 1 Examples and practice exercises.



- Consider a sphere of radius a centred at the origin. Let the volume be filled uniformly with a volume charge density ($C/\text{cubic m}$).
- The charge associated with the elemental volume chosen at is,
- The total charge in this sphere is,



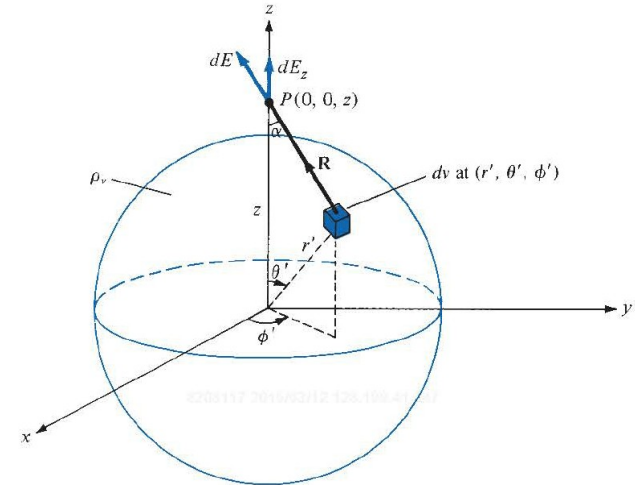
EMF THEORY – Unit - 1

ELECTRIC FIELD DUE TO A VOLUME CHARGE



-
- The proof can be done in Gauss law applications. Here Formula to be known no derivation.

re Numerical. Text 1 Examples and practice exercises.



- This is given by . Now the electric flux in terms of is given by;
- The vector field is called **Electric Flux Density (C/sq. m)**, is also known as **Electric Displacement**. The flux is in **Coulombs**.
- Since , we can get the expressions for if is known for various charge distributions.
- - due to Point charge

EMF THEORY – Unit - 1

ELECTRIC FLUX DENSITY



- - due to Infinite Surface charge
- - due to Infinite Line charge.
- - due to Volume charge distribution.
- From the above expressions, it is seen that
is a function of the charge (distribution)
and position only.
- Numerical !

ELECTRO MAGNETI C THEORY

El ect r i c Fl ux Densi t y



$$\mathbf{D} = \epsilon_0 \mathbf{a}_r$$

ELECTRO MAGNETI C

THEORY

Numerical Examples on Electric Flux Density and Electric Field

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ELECTRO MAGNETIC THEORY

Numerical Example



Determine $\mathbf{D}$ at $(4, 0, 3)$ if there is a point charge -5π mC at $(4, 0, 0)$ and a line charge 3π mC/m along the y-axis.

Solution:

Let $\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L$ where $\mathbf{D}_Q$ and $\mathbf{D}_L$ are flux densities due to the point charge and line charge, respectively, as shown in Figure 4.11:

$$\mathbf{D}_Q = \epsilon_0 \mathbf{E} = \frac{Q}{4\pi R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi |\mathbf{r} - \mathbf{r}'|^3}$$

where $\mathbf{r} - \mathbf{r}' = (4, 0, 3) - (4, 0, 0) = (0, 0, 3)$. Hence,

$$\mathbf{D}_Q = \frac{-5\pi \cdot 10^{-3}(0, 0, 3)}{4\pi |(0, 0, 3)|^3} = -0.138 \mathbf{a}_z \text{ mC/m}^2$$

Also

$$\mathbf{D}_L = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho$$

In this case

$$\mathbf{a}_\rho = \frac{(4, 0, 3) - (0, 0, 0)}{|(4, 0, 3) - (0, 0, 0)|} = \frac{(4, 0, 3)}{5}$$

$$\rho = |(4, 0, 3) - (0, 0, 0)| = 5$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd.,

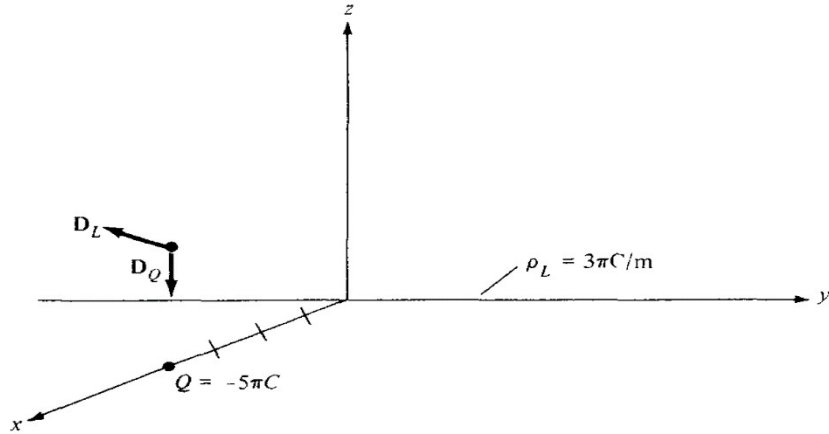


Figure 4.11 Flux density **D** due to a point charge and an infinite line charge.

Hence,

$$\mathbf{D}_L = \frac{3\pi}{2\pi(25)} (4\mathbf{a}_x + 3\mathbf{a}_z) = 0.24\mathbf{a}_x + 0.18\mathbf{a}_z \text{ mC/m}^2$$

Thus

$$\begin{aligned} \mathbf{D} &= \mathbf{D}_Q + \mathbf{D}_L \\ &= 240\mathbf{a}_x + 42\mathbf{a}_z \text{ } \mu\text{C/m}^2 \end{aligned}$$

ELECTRO MAGNETIC THEORY

Merical Example Cont d. ,



Given that $\mathbf{D} = z\rho \cos^2\phi \mathbf{a}_z \text{ C/m}^2$, calculate the charge density at $(1, \pi/4, 3)$ and the total charge enclosed by the cylinder of radius 1 m with $-2 \leq z \leq 2$ m.

ELECTRO MAGNETIC THEORY

Merical Example Cont d. ,



$$\rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_z}{\partial z} = \rho \cos^2 \phi$$

At $(1, \pi/4, 3)$, $\rho_v = 1 \cdot \cos^2(\pi/4) = 0.5 \text{ C/m}^3$. The total charge enclosed by the cylinder can be found in two different ways.

ELECTRO MAGNETIC THEORY

Merical Example Contd.,

Method 1: This method is based directly on the definition of the total volume charge.

$$\begin{aligned} Q &= \int_v \rho_v dv = \int_v \rho \cos^2 \phi \rho d\phi d\rho dz \\ &= \int_{z=-2}^2 dz \int_{\phi=0}^{2\pi} \cos^2 \phi d\phi \int_{\rho=0}^1 \rho^2 d\rho = 4(\pi)(1/3) \\ &= \frac{4\pi}{3} \text{ C} \end{aligned}$$

Method 2: Alternatively, we can use Gauss's law.

$$\begin{aligned} Q &= \Psi = \oint \mathbf{D} \cdot d\mathbf{S} = \left[\int_s + \int_t + \int_b \right] \mathbf{D} \cdot d\mathbf{S} \\ &= \Psi_s + \Psi_t + \Psi_b \end{aligned}$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd.,

$\Psi_s = 0$, for $\Psi_r, d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$ so

$$\begin{aligned}\Psi_t &= \int_{\rho=0}^1 \int_{\phi=0}^{2\pi} z\rho \cos^2 \phi \rho d\phi d\rho \Big|_{z=2} = 2 \int_0^1 \rho^2 d\rho \int_0^{2\pi} \cos^2 \phi d\phi \\ &= 2 \left(\frac{1}{3} \right) \pi = \frac{2\pi}{3}\end{aligned}$$

and for $\Psi_b, d\mathbf{S} = -\rho d\phi d\rho \mathbf{a}_z$, so

$$\begin{aligned}\Psi_b &= - \int_{\rho=0}^1 \int_{\phi=0}^{2\pi} z\rho \cos^2 \phi \rho d\phi d\rho \Big|_{z=-2} = 2 \int_0^1 \rho^2 d\rho \int_0^{2\pi} \cos^2 \phi d\phi \\ &= \frac{2\pi}{3}\end{aligned}$$

Thus

$$Q = \Psi = 0 + \frac{2\pi}{3} + \frac{2\pi}{3} = \frac{4\pi}{3} \text{ C}$$

as obtained previously.

ELECTRO MAGNETIC THEORY

Merical Example Cont d. ,

2. A charge distribution with spherical symmetry has density

$$\rho_v = \begin{cases} \frac{\rho_0 r}{R}, & 0 \leq r \leq R \\ 0, & r > R \end{cases}$$

Determine **E** everywhere.

ELECTRO MAGNETIC THEORY

Merical Example Contd.,

2. The charge distribution is similar to that in Figure 4.16. Since symmetry exists, we can apply Gauss's law to find $\mathbf{E}$.

$$\epsilon_o \oint \mathbf{E} \cdot d\mathbf{S} = Q_{\text{enc}} = \int \rho_v dv$$

(a) For $r < R$

$$\begin{aligned}\epsilon_o E_r 4\pi r^2 &= Q_{\text{enc}} = \int_0^r \int_0^\pi \int_0^{2\pi} \rho_v r^2 \sin \theta d\phi d\theta dr \\ &= \int_0^r 4\pi r^2 \frac{\rho_o r}{R} dr = \frac{\rho_o \pi r^4}{R}\end{aligned}$$

or

$$\mathbf{E} = \frac{\rho_o r^2}{4\epsilon_o R} \mathbf{a}_r$$

(b) For $r > R$,

$$\begin{aligned}\epsilon_o E_r 4\pi r^2 &= Q_{\text{enc}} = \int_0^R \int_0^\pi \int_0^{2\pi} \rho_v r^2 \sin \theta d\phi d\theta dr \\ &= \int_0^R \frac{\rho_o r}{R} 4\pi r^2 dr + \int_R^r 0 \cdot 4\pi r^2 dr \\ &= \pi \rho_o R^3\end{aligned}$$

or

$$\mathbf{E} = \frac{\rho_o R^3}{4\epsilon_o r^2} \mathbf{a}_r$$

ELECTRO MAGNETIC THEORY

Practical Example Contd., (Assignment)

1. A point charge of 30 nC is located at the origin while plane $y = 3$ carries charge 10 nC/m^2 . Find $\mathbf{D}$ at $(0, 4, 3)$.
2. If $\mathbf{D} = (2y^2 + z)\mathbf{a}_x + 4xy\mathbf{a}_y + x\mathbf{a}_z \text{ C/m}^2$, find
 - (a) The volume charge density at $(-1, 0, 3)$
 - (b) The flux through the cube defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$
 - (c) The total charge enclosed by the cube
3. A charge distribution in free space has $\rho_v = 2r \text{ nC/m}^3$ for $0 \leq r \leq 10 \text{ m}$ and zero otherwise. Determine $\mathbf{E}$ at $r = 2 \text{ m}$ and $r = 12 \text{ m}$.

ELECTRO MAGNETIC THEORY

cal Example Contd., (Assignment answers)



1. **Answer:** $5.076\mathbf{a}_y + 0.0573\mathbf{a}_z$ nC/m².

2. **Answer:** (a) -4 C/m³, (b) 2 C, (c) 2 C.

3. $226\mathbf{a}_r$ V/m, $3.927\mathbf{a}_r$ kV/m.

ELECTRO MAGNETIC THEORY

Gauss Law



Gauss Law states that the total electric flux Ψ through any Closed surface is equal to the total charge enclosed by that surface. Thus $\Psi = Q_{enc}$

= Total Charge enclosed $Q =$

$Q =$

EMF THEORY – Unit - 1

DIVERGENCE OF A VECTOR



- - Cartesian
- - Cylindrical
- - Spherical
- Properties of Divergence of a Vector Field;
 - It produces a scalar field due to 'dot' product.

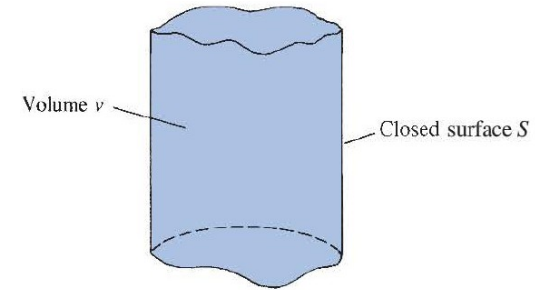
ELECTRO MAGNETIC THEORY



Divergence

- The **divergence** of $\vec{E}$ at a given point P is the outward flux per unit volume as the volume shrinks at P .
-
- where V is the volume enclosed by the closed surface S in which P is located.

- Recall
- From the above equation, we can say;
- - Divergence Theorem or Gauss – Otrogradsky Theorem
- The **Divergence Theorem** states that the total outward flux of a vector field through the closed surface **S** is the same as the volume integral of the divergence of .
- To determine the flux of through a closed surface, we just need to find



ELECTRO MAGNETIC THEORY

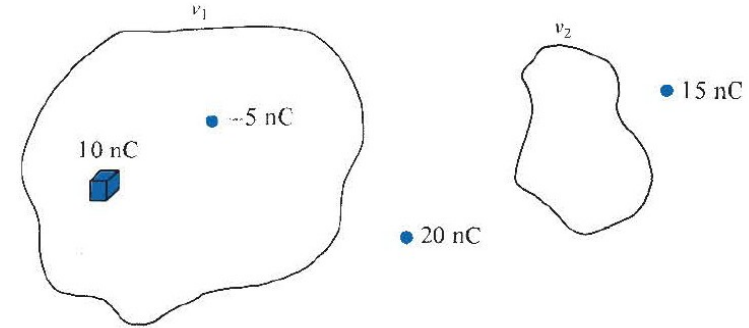
Gauss Divergence Theorem (Maxwell's
Equation)

By applying Divergence Theorem, We get

EMF Theory – Unit - 1

GAUSS' S LAW AND MAXWELL' S EQUATION

- ~~and are closed surfaces.~~ The total flux leaving is $10 - 5 = 5$ nC because only 10 nC and -5 nC are enclosed by .
- Although the charges 20 nC and 15 nC outside do contribute to the flux crossing , the **net flux crossing** , according to Gauss's Law, is irrespective of those charges outside .
- Similarly, the total flux leaving is zero, since it does not enclose any charge inside it.



EMF Theory – Unit - 1

APPLICATIONS OF GAUSS' S LAW

▪ Preliminaries:

- Check whether a symmetry exists to compute or .
- If symmetric charge distribution exists, construct a mathematical closed surface known as Gaussian Surface.
- The surface is so chosen such that $\vec{E}$ is normal or tangential to the surface.
- When $\vec{E}$ is tangential to the surface
- When $\vec{E}$ is normal to the surface, , since $\vec{E}$ is perpendicular to the surface.



ELECTRO MAGNETI C THEORY

Gauss Law appl i cat i ons



Poi nt Char ge

I nf i ni t e Li ne Char ge

I nf i ni t e Sheet Char ge

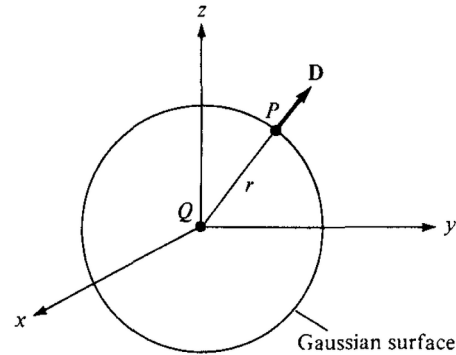
Uni f or m l y char ged Spher es

ELECTRO MAGNETI C THEORY

Gauss Law appl i cat i ons t o poi nt charge

$$Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_r \oint dS = D_r 4\pi r^2$$

$$\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$$



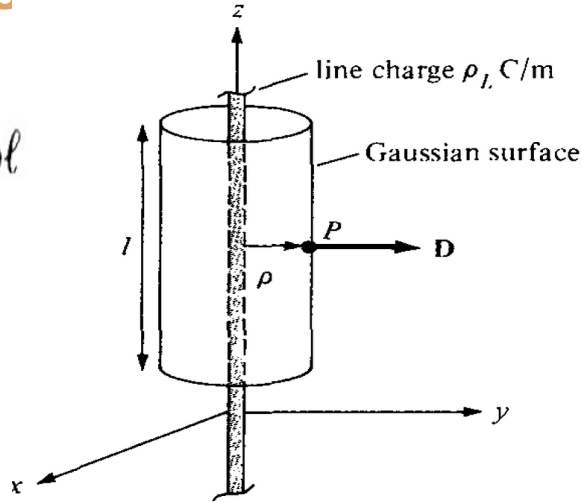
ELECTRO MAGNETI C THEORY

Gauss Law appl i cat i ons t o

I n f i n i t e l i n e c h a r g e

$$\rho_L \ell = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_\rho \oint dS = D_\rho 2\pi\rho\ell$$

$$\mathbf{D} = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho$$



ELECTRO MAGNETIC THEORY

Gauss Law applications to Infinite Sheet charge

$$\rho_s \int dS = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_z \left[\int_{\text{top}} dS + \int_{\text{bottom}} dS \right]$$

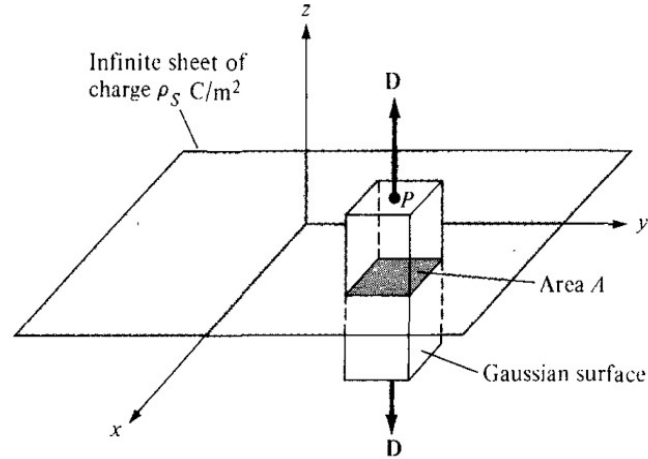
$$\rho_s A = D_z (A + A)$$

and thus

$$\mathbf{D} = \frac{\rho_s}{2} \mathbf{a}_z$$

or

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z$$



ELECTRO MAGNETIC THEORY

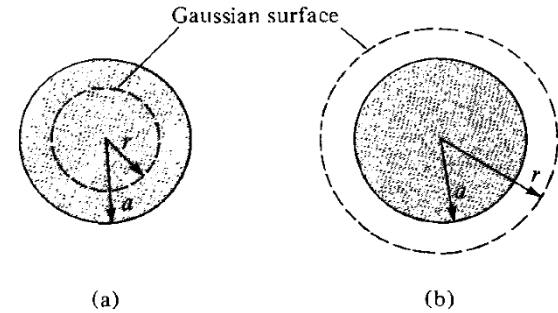
Gauss Law applications to Uniformly Charged Sphere

For $r \leq a$, the total charge enclosed by the spherical surface of radius r , as shown in Figure, is

$$\begin{aligned} Q_{\text{enc}} &= \int \rho_v dv = \rho_v \int dv = \rho_v \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^r r^2 \sin \theta dr d\theta d\phi \\ &= \rho_v \frac{4}{3} \pi r^3 \end{aligned}$$

and

$$\begin{aligned} \Psi &= \oint \mathbf{D} \cdot d\mathbf{S} = D_r \oint dS = D_r \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin \theta d\theta d\phi \\ &= D_r 4\pi r^2 \end{aligned}$$



ELECTRO MAGNETIC THEORY

Gauss Law applications to Uniformly Charged Sphere

$$D_r 4\pi r^2 = \frac{4}{3} \pi a^3 \rho_v$$

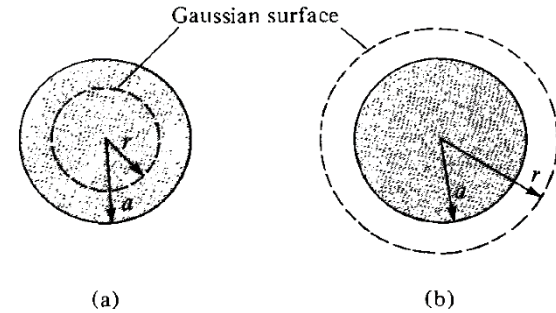
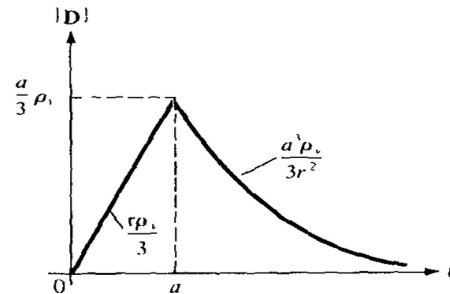
or

$$\mathbf{D} = \frac{a^3}{3r^2} \rho_v \mathbf{a}_r \quad r \geq a$$

Thus from eqs. $\mathbf{D}$ everywhere is given by

$$\mathbf{D} = \begin{cases} \frac{r}{3} \rho_v \mathbf{a}_r & 0 < r \leq a \\ \frac{a^3}{3r^2} \rho_v \mathbf{a}_r & r \geq a \end{cases}$$

and $|\mathbf{D}|$ is as sketched in Figure



ELECTRO MAGNETI C F I ELD

THEORY

El e c t r i c P o t e n t i a l

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EMF Theory – Unit - 1

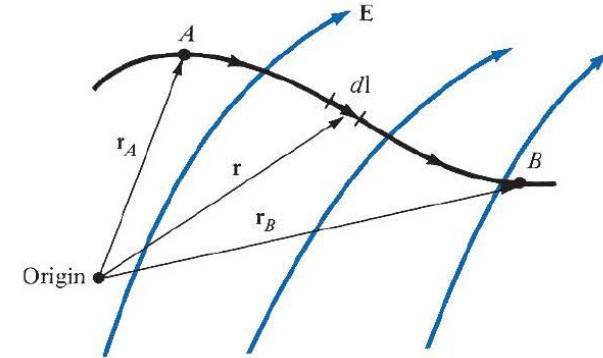
ELECTRIC POTENTIAL

- From Coulomb's Law, the force on q is $F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2}$, so that the work done (Force $\times$ Distance) in displacing the charge by dr is $dW = F dr$.

- The negative sign indicates that the work is being done by an external agent.
- Thus, the total work done, or the potential energy required, in moving from A to B is;

- A Scalar

- Dividing by q in the above eqn. gives the potential difference between A and B as

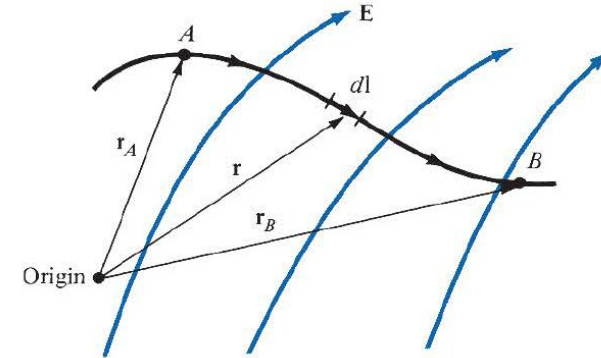


EMF Theory – Unit - 1

ELECTRIC POTENTIAL

- This quantity, denoted as V is known as the “**potential difference**” between points A and B.

- Thus;
- In determining V , A is the initial point while B is the final point.
- If V is **negative**, there is a **loss** in potential energy in moving from A to B; this implies that the work is being done by the field.
- If V is positive, there is a gain in the potential energy in the movement; an



EMF Theory – Unit - 1

~~ELECTRIC POTENTIAL~~ Definition

The potential at any point is the potential difference between that point and a chosen point (or reference point) at which the potential is zero.

- In other words, if one assumes zero potential at infinity, the potential at a distance 'r' from the point charge is the work done per unit charge by an external agent in transferring a test charge from infinity to that point.
- Thus,
- If the point charge in Eqn. 1 is not located at origin, but at a point whose position vector is $\vec{r}$, the potential or



EMF Theory – Unit - 1

ELECTRIC POTENTIAL



- Having considered the potential due to a point charge, the logic can be extended to other types of charge distribution because any charge distribution can be regarded as consisting of point charges !
- **Superposition Principle** can also be applied to electric potential like electric fields !
- For ‘ n ’ point charges located at points with position vectors , the potential at a point **P** with a position vector is given by-

EMF Theory – Unit - 1

ELECTRIC POTENTIAL



- - Point charges
- For other charge distributions;
- - Line charge
- - Surface charge
- - Volume charge
- (') indicates source point. Unprimed coordinates refer to the field point at which is to be determined.

EMF Theory – Unit - 1

~~ELECTRIC POTENTIAL~~

- While obtaining the expression and all other expressions, the zero potential (reference) point has been chosen arbitrarily to be at infinity.
- If any other point is chosen as reference, then the expression becomes , where k is a constant that is to be determined at the chosen point of reference. Same analogy applies to other expressions as well.
- The potential at any point can be evaluated in two ways;
- Method 1: If the charge distribution is known we can use any one of the



EMF Theory – Unit - 1

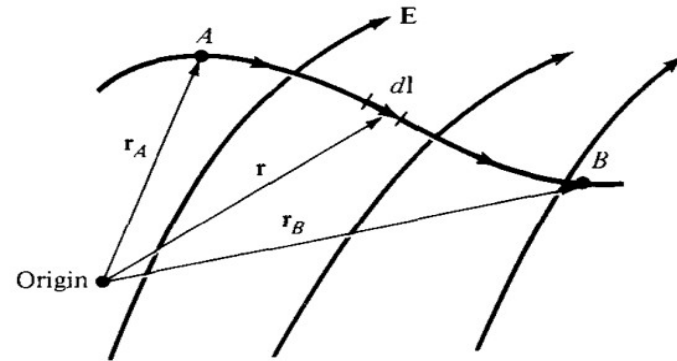
ELECTRIC POTENTIAL



- Method 2: If $\vec{E}$ is known, we can use
- The potential difference can be generally found from
- Numericals

ELECTRO MAGNETI C THEORY

El ect r i c Pot ent i al



$$= - \int_A^B \mathbf{E} \cdot d\mathbf{l}$$

ELECTRO MAGNETIC THEORY

Electric potential for a point charge derived using E

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

$$\begin{aligned} V_{AB} &= - \int_{r_A}^{r_B} \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \cdot d\mathbf{r} \mathbf{a}_r \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \end{aligned}$$

$$V = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l}$$

$$V(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|}$$

or

$$V_{AB} = V_B - V_A$$

ELECTRO MAGNETIC THEORY

Numerical Examples on Electric

Two point charges $-4 \mu\text{C}$ and $5 \mu\text{C}$ are located at $(2, -1, 3)$ and $(0, 4, -2)$, respectively. Find the potential at $(1, 0, 1)$ assuming zero potential at infinity.

Solution:

Let

$$Q_1 = -4 \mu\text{C}, \quad Q_2 = 5 \mu\text{C}$$

$$V(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_1|} + \frac{Q_2}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_2|} + C_0$$

If $V(\infty) = 0$, $C_0 = 0$,

$$|\mathbf{r} - \mathbf{r}_1| = |(1, 0, 1) - (2, -1, 3)| = |(-1, 1, -2)| = \sqrt{6}$$

$$|\mathbf{r} - \mathbf{r}_2| = |(1, 0, 1) - (0, 4, -2)| = |(1, -4, 3)| = \sqrt{26}$$

Hence

$$\begin{aligned} V(1, 0, 1) &= \frac{10^{-6}}{4\pi \times \frac{10^{-9}}{36\pi}} \left[\frac{-4}{\sqrt{6}} + \frac{5}{\sqrt{26}} \right] \\ &= 9 \times 10^3 (-1.633 + 0.9806) \\ &= -5.872 \text{ kV} \end{aligned}$$

ELECTRO MAGNETIC THEORY

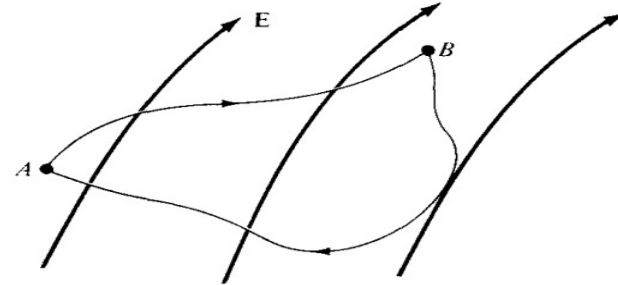
Relation Between E and V

$$V_{BA} = -V_{AB}$$

that is, $V_{BA} + V_{AB} = \oint \mathbf{E} \cdot d\mathbf{l} = 0$

or

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$



ELECTRO MAGNETIC THEORY



Relation Between E and V

$$\oint \mathbf{E} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = 0$$

or

$$\nabla \times \mathbf{E} = 0$$

From the way we defined potential, $V = -\int \mathbf{E} \cdot d\mathbf{l}$, it follows that

$$dV = -\mathbf{E} \cdot d\mathbf{l} = -E_x dx - E_y dy - E_z dz$$

ELECTRO MAGNETIC THEORY

Relation Between E and V

But

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz$$

Comparing the two expressions for dV , we obtain

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}$$

Thus:

$$\mathbf{E} = -\nabla V$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd. ,



Given the potential $V = \frac{10}{r^2} \sin \theta \cos \phi$,

(a) Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.

(b) Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.

Solution:

(a) $\mathbf{D} = \epsilon_0 \mathbf{E}$

But

$$\begin{aligned}\mathbf{E} &= -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \right] \\ &= \frac{20}{r^3} \sin \theta \cos \phi \mathbf{a}_r - \frac{10}{r^3} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{10}{r^3} \sin \phi \mathbf{a}_\phi\end{aligned}$$

or

$$W = \frac{45}{16} Q = 28.125 \mu\text{J}$$

Method 2:

Since V is known, this method is a lot easier.

$$\begin{aligned} W &= -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} = QV_{AB} \\ &= Q(V_B - V_A) \\ &= 10 \left(\frac{10}{16} \sin 90^\circ \cos 60^\circ - \frac{10}{1} \sin 30^\circ \cos 120^\circ \right) \cdot 10^{-6} \\ &= 10 \left(\frac{10}{32} - \frac{-5}{2} \right) \cdot 10^{-6} \\ &= 28.125 \mu\text{J as obtained before} \end{aligned}$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd.,



Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4, 3, 6)$, find V , E , direction of E , D , and ρ_v .

$$V_p = 2(-4)^2(3) - 5(6) = \underline{\underline{66 \text{ V}}}$$

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right) = -4xy\mathbf{a}_x - 2x^2\mathbf{a}_y + 5\mathbf{a}_z$$

$$\mathbf{E}_p = -4(-4)(3)\mathbf{a}_x - 2(-4)^2\mathbf{a}_y + 5\mathbf{a}_z = \underline{\underline{48\mathbf{a}_x - 32\mathbf{a}_y + 5\mathbf{a}_z \text{ V/m}}}$$

$$\mathbf{a}_{E,p} = \frac{\mathbf{E}_p}{\|\mathbf{E}_p\|}$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd.,



$$\mathbf{D}_P = \epsilon_0 \mathbf{E}_P = \underline{\underline{425\mathbf{a}_x - 283.3\mathbf{a}_y + 44.27\mathbf{a}_z \text{ pC/m}^3}}$$

$$\rho_v = \text{div } \mathbf{D} = \text{div } \epsilon_0 \mathbf{E} = (8.854 \times 10^{-12})(-4y) = -35.42y \text{ pC/m}^3$$

$$\text{At } P, \rho_v = -35.42(3) \text{ pC/m}^3 = \underline{\underline{-106.26 \text{ pC/m}^3}}$$

ELECTRO MAGNETIC

THEORY Maxwell's Equations

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ELECTRO MAGNETI C THEORY

Maxwell I ' s Equat i ons



Maxwell I ' s Equat i on

I

$$\text{div } \mathbf{D} = \rho_v$$

Maxwell I ' s Equat i on II

ELECTRO MAGNETIC THEORY

Numerical Example



Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4, 3, 6)$, find the following: potential V , electric field intensity E at point P .

$$\begin{aligned} V_P &= 2(-4)^2(3) - 5(6) = 66 \text{ V} \\ E &= -4xya_x - 2x^2a_y + 5a_z \\ E_P &= -48a_x - 32a_y + 5a_z \end{aligned}$$

ELECTRO MAGNETIC THEORY

numerical Example Contd. ,



Let $D = y^2z^3 a_x + 2xyz^3 a_y + 3xy^2z^2 a_z$ pC/m² in free space.

(a) Find the total electric flux passing through the surface $x = 3$, $0 \leq y \leq 2$, $0 \leq z \leq 1$ in a direction away from the origin.

(b) Find $|E|$ at $P(3, 2, 1)$.

(c) Find the total charge contained in an incremental sphere having a radius of $2 \mu\text{m}$.

ELECTRO MAGNETIC THEORY

Numerical Example Contd. ,



$$\begin{aligned} \text{a) } \psi &= \int_S \mathbf{D}_S \cdot d\mathbf{S} \\ &= \int_{z=0}^1 \int_{y=0}^2 (y^2 z^3 \mathbf{a}_x + 2xyz^3 \mathbf{a}_y + 3xy^2 z^2 \mathbf{a}_z) \cdot (dydz \mathbf{a}_x) \Big|_{x=3} \\ &= \int_0^1 \int_0^2 y^2 z^3 dy dz \\ &= \frac{1}{3} y^3 \Big|_0^2 \frac{1}{4} z^4 \Big|_0^1 \\ &= \underline{\underline{\frac{2}{3} \text{ pC}}} \end{aligned}$$

ELECTRO MAGNETIC THEORY

Numerical Example Contd.,



$$b) \quad \mathbf{D} = y^2 z^3 \mathbf{a}_x + 2xyz^3 \mathbf{a}_y + 3xy^2 z^2 \mathbf{a}_z$$

$$\begin{aligned} \mathbf{D}_P &= (2)^2 (1)^3 \mathbf{a}_x + 2(3)(2)(1)^3 \mathbf{a}_y + 3(3)(2)^2 (1)^2 \mathbf{a}_z \\ &= 4\mathbf{a}_x + 12\mathbf{a}_y + 36\mathbf{a}_z \text{ pC/m}^2 \end{aligned}$$

$$|\mathbf{D}_P| = D_P = \sqrt{(4)^2 + (12)^2 + (36)^2}$$

$$|\mathbf{E}_P| = \frac{|\mathbf{D}_P|}{\epsilon_0}$$

$$= \frac{38.158 \text{ pC/m}^2}{8.854 \times 10^{-12}} = \underline{\underline{4.31 \text{ V/m}}}$$

ELECTRO MAGNETI C FI ELD THEORY

El ect r i c Di pol e

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ELECTRO MAGNETIC FIELD THEORY

Electric Dipole Definition



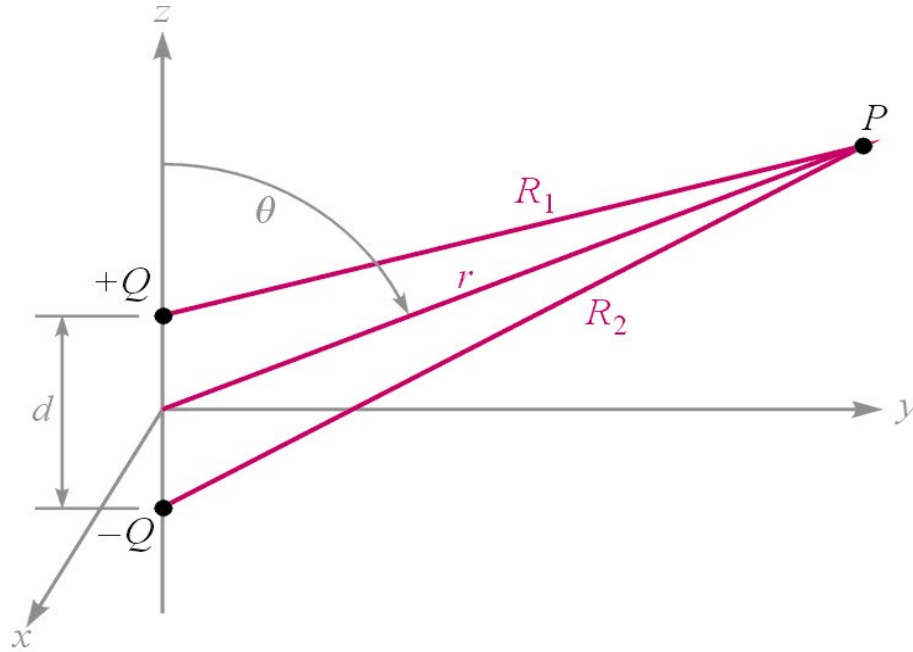
An electric dipole, or simply a dipole is

The two point charges of equal magnitude but opposite sign.

They are separated by a very small distance compared to a point at which electric field to be found.

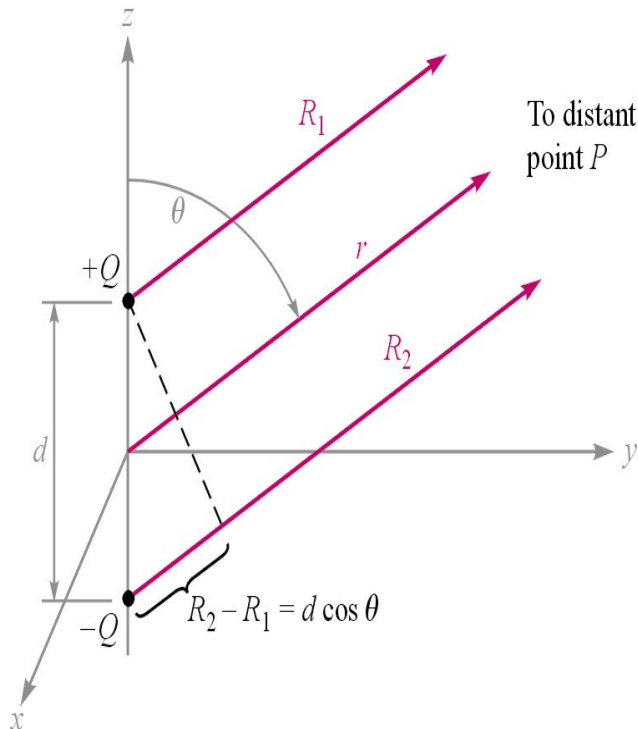
ELECTRO MAGNETIC FIELD THEORY

Electric Dipole



ELECTRO MAGNETIC FIELD THEORY

Electric Dipole



To approximate the potential of a dipole, assume R1 and R2 are parallel since the point P is very distant

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$V = \frac{Q \cdot d \cdot \cos(\theta)}{4\pi\epsilon_0 r^2}$$

ELECTRO MAGNETIC FIELD THEORY

Dipole Moment



The dipole moment is assigned the symbol p and is equal to the product of charge and separation

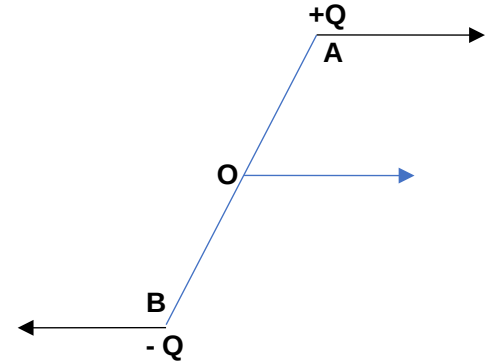
$$p = Q \cdot d$$

The dipole moment expression simplifies the potential field equation

EMF Theory – Unit - 1

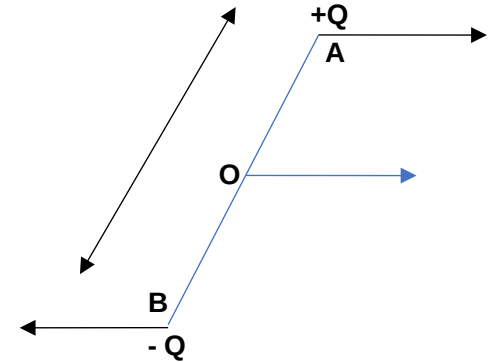
TORQUE DUE TO ELECTRIC FIELD ON A DIPOLE

- Consider a dipole placed in a uniform electric field. The dipole consists of two charges and, placed at **A** and **B**, respectively, as shown.
- Let the distance between the two charges and the midpoint of the distance vector is at origin.
- Assume that the axis of the dipole makes an angle with .
- The forces on and are and
- The two forces create a ‘couple’ and

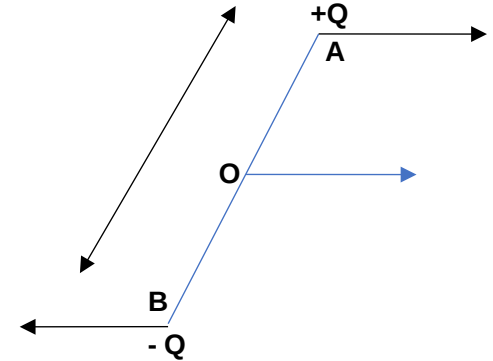


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- The torque produced by each of the two forces is given by the product of the magnitude of any one force and the perpendicular distance from the axis of rotation of the force.
- The net torque, is then, given by:
- Hypotenuse = . The multiplier **2** appears in the expression since the torque is generated by two forces.
- Thus the magnitude of the torque acting on a dipole is the product of the magnitude of the force or and the



- Therefore;
- as is same as – dipole moment
- Hence , where is the dipole moment !
- The direction of the torque will always be perpendicular to the plane containing the dipole and .
- In this case, the torque direction is into the plane of the paper at the origin.



ELECTRO MAGNETIC THEORY

Numerical Example on Dipole Moment and Electric Potential



An electric dipole located at the origin in free space has a moment $\mathbf{p} = 3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z$ nC·m. Find V at the points $(2, 3, 4)$ and $(2.5, 30^\circ, 40^\circ)$.

ELECTRO MAGNETIC THEORY

Numerical Example



$$\mathbf{p} := \begin{pmatrix} 3 \cdot 10^{-9} \\ -2 \cdot 10^{-9} \\ 1 \cdot 10^{-9} \end{pmatrix}$$

$$\epsilon_0 := 8.854 \cdot 10^{-12}$$

$$\mathbf{P} := \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

$$\mathbf{V} := \frac{\mathbf{p}}{4 \pi \cdot \epsilon_0 \cdot (|\mathbf{P}|)^2} \cdot \frac{\mathbf{P}}{|\mathbf{P}|}$$

$$V \equiv 0.23$$

$$\mathbf{P}_{\text{spherical}} := \begin{pmatrix} 2.5 \\ 30 \cdot \frac{\pi}{180} \\ 40 \cdot \frac{\pi}{180} \end{pmatrix}$$

Transform this
into rectangular
coordinates

$$\mathbf{P}_{\text{rectangular}} := \begin{pmatrix} 2.5 \cdot \sin\left(30 \cdot \frac{\pi}{180}\right) \cdot \cos\left(40 \cdot \frac{\pi}{180}\right) \\ 2.5 \cdot \sin\left(30 \cdot \frac{\pi}{180}\right) \cdot \sin\left(40 \cdot \frac{\pi}{180}\right) \\ 2.5 \cdot \cos\left(30 \cdot \frac{\pi}{180}\right) \end{pmatrix}$$

$$\mathbf{P}_{\text{rectangular}} \equiv \begin{pmatrix} 0.958 \\ 0.803 \\ 2.165 \end{pmatrix}$$

ELECTRO MAGNETI C

THEORY

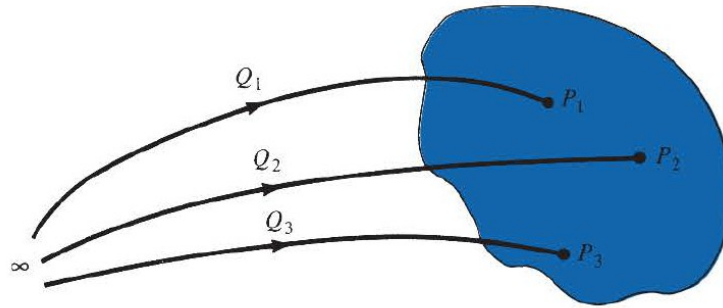
Energy and Energy Density

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EMF Theory – Unit - 1

ENERGY DENSITY IN ELECTROSTATIC FIELDS



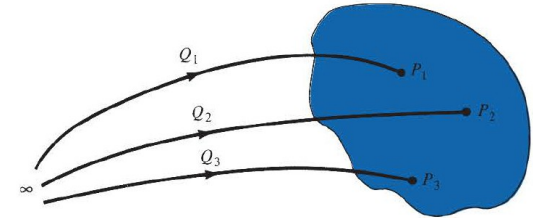
- To determine the energy present in an assembly of charges, we must first determine the amount of work necessary to assemble them
- Suppose we wish to place / position 3 point charges , and in an **initially empty space** as shown.

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ENERGY DENSITY IN ELECTROSTATIC FIELDS

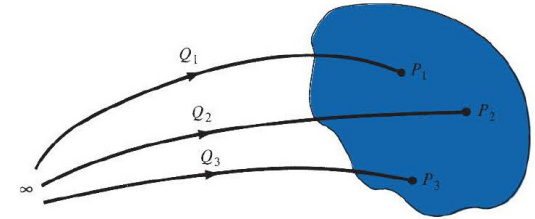


- No work is required to transfer from infinity to because the space is initially charge free and there is no electric field !
- The work done in transferring from infinity to is equal to the product of and the potential at due to .
- Similarly, the work done in positioning at is equal to , where and are the potentials at due to and , respectively.



- Hence the total work done in positioning

- If the charges were positioned in the reverse order;
- , where V is the potential at P due to Q , and V_1, V_2 are, respectively, the potentials at P due to Q_1 and Q_2 .
- Adding both of the above expressions gives;

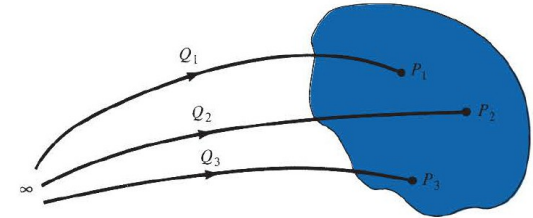


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ENERGY DENSITY IN ELECTROSTATIC FIELDS



- Where V_1 and V_2 are the potentials at P_1 and P_2 , respectively.
- In general, if there are point charges;
- $W = \int_{\infty}^{P_1} \mathbf{E} \cdot d\mathbf{l}$ In Joules.
- Instead of point charges, the region has a continuous charge distribution, the summation becomes integration.
- (line charge), (Surface charge), (volume charge)



EMF Theory – Unit - 1

ENERGY DENSITY IN ELECTROSTATIC FIELDS



- Consider the volume charge equation. From Maxwell's Equation, we have;
- 1
- For any vector and a scalar, following identities hold;
- or
- Substituting the above in Eqn. 1, we have;
- Applying Divergence Theorem to the first

- From previous discussions, ϕ varies as $\frac{1}{r}$ and E as for point charges; ϕ varies as $\frac{1}{r^2}$ and E as for dipoles and so on.
- Hence ϕ in the first term of RHS must vary at least while E varies as $\frac{1}{r^2}$. Consequently, the first integral must tend to zero as the surface becomes large.
- The expression for U reduces to

EMF Theory – Unit - 1

ENERGY DENSITY IN ELECTROSTATIC FIELDS



- Since , the electrostatic energy is given by;
- Then the Electrostatic Energy density (Joules / cu. m) is given by;
- while
- Numerical s.

ELECTRO MAGNETIC THEORY

Energy Concepts



- 1 nC, 4 nC, and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$ and $(1, 0, 0)$ respectively.

Find the energy in the system

ELECTRO MAGNETIC THEORY

Energy Concepts Contd. ,

$$\begin{aligned} W &= \frac{1}{2} \sum_{k=1}^3 Q_k V_k = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3) \\ &= \frac{Q_1}{2} \left[\frac{Q_2}{4\pi\epsilon_0(1)} + \frac{Q_3}{4\pi\epsilon_0(1)} \right] + \frac{Q_2}{2} \left[\frac{Q_1}{4\pi\epsilon_0(1)} + \frac{Q_3}{4\pi\epsilon_0(\sqrt{2})} \right] \\ &\quad + \frac{Q_3}{2} \left[\frac{Q_1}{4\pi\epsilon_0(1)} + \frac{Q_2}{4\pi\epsilon_0(\sqrt{2})} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right) \\ &= 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ} \end{aligned}$$

ELECTRO MAGNETIC FIELD THEORY

Assignment



1. Point charges $Q_1 = 1 \text{ nC}$, $Q_2 = -2 \text{ nC}$, $Q_3 = 3 \text{ nC}$, and $Q_4 = -4 \text{ nC}$ are positioned one at a time and in that order at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 0, -1)$ and $(0, 0, 1)$ respectively.

Calculate the energy in the system after each charge is positioned.

ELECTRO MAGNETIC FIELD THEORY

Assignment



Answer: 0, -1.8 nJ, -29.18 nJ, -68.27 nJ



THANK YOU

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